

- 8 There is considerable variation in the ecosystems that occur in the continent of North America. These include coniferous forest, prairie grassland, scrub and desert. Large areas of land that once contained natural ecosystems are now used for agriculture.

(a) Explain how the variation in ecosystems in North America contributes to biodiversity.

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..... [4]

- (b) The diversity of some beetle species that feed on animal dung (faeces) was investigated at two types of grassland site in North America. The first type of grassland site was grazed by cattle and the second type of site was not grazed.

Dung beetles were collected, identified and counted from two areas of the same total size. Some of the results are shown in Table 8.1.

Table 8.1

beetle species	number of dung beetles on grassland grazed by cattle	number of dung beetles on grassland not grazed
<i>Onthophagus pennsylvanicus</i>	4267	6641
<i>Canthon ebenus</i>	2005	774
<i>Canthon pilularius</i>	353	108
<i>Onthophagus hecate</i>	218	85
total	6843	7608

- (i) State the null hypothesis for a statistical test comparing the data from the two types of site.

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 [1]

- (ii) State how many genera and how many species of beetle are shown in Table 8.1.

genera species [1]

- (iii) Simpson's Index of Diversity for the beetles on the grassland grazed by cattle was calculated as 0.521 using the formula:

$$D = 1 - \left(\sum \left(\frac{n}{N} \right)^2 \right)$$

n = number of individuals of each species present in the sample
 N = the total number of all individuals of all species

Calculate Simpson's Index of Diversity for the beetles on the grassland that was not grazed. Complete Table 8.2 to show your working. Show all working to **three** decimal places. Write your final answer on the dotted line.

Table 8.2

species	number on grassland not grazed	$\frac{n}{N}$	$\left(\frac{n}{N} \right)^2$
<i>Onthophagus pennsylvanicus</i>	6641		
<i>Canthon ebenus</i>	774		
<i>Canthon pilularius</i>	108		
<i>Onthophagus hecate</i>	85		
total	7608		

Simpson's Index of Diversity = [3]

- (iv) Describe what the results in Table 8.1 **and** both figures for Simpson's Index of Diversity show about the effect of grazing on the diversity of dung beetles.

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 [2]

Section B

Answer **one** question.